Inverse & Composite Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

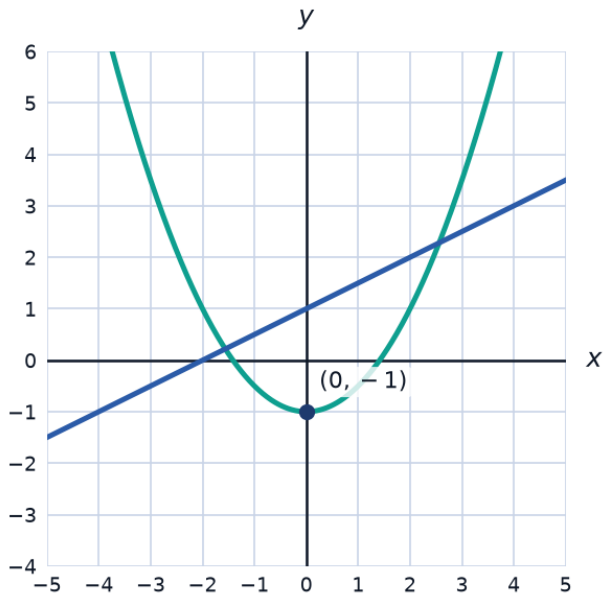
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Composing functions

Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

2 [Foundational] • Target: Composing functions

Given $f(x) = \sqrt{x + 4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(2)$ and state the domain restriction that makes $f \circ g$ defined.



3 [Foundational] • Target: Decomposing a composite function

Write $h(x) = (3x - 5)^4$ as a composition $h(x) = f(g(x))$ for two simpler functions f and g.

4 [Foundational] • Target: Decomposing a composite function

Write $h(x) = \sqrt{x^2 + 1}$ as a composition of two functions.

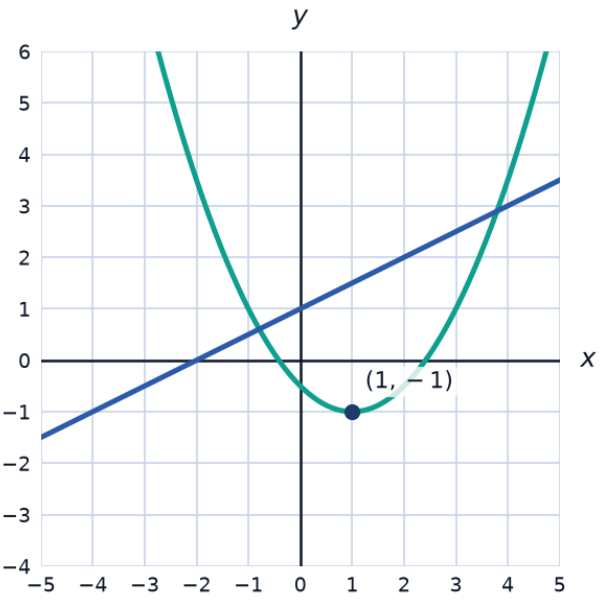
5 [Foundational] • Target: Verifying inverse functions

Verify algebraically that $f(x) = \frac{x - 2}{3}$ and $g(x) = 3x + 2$ are inverses of each other.

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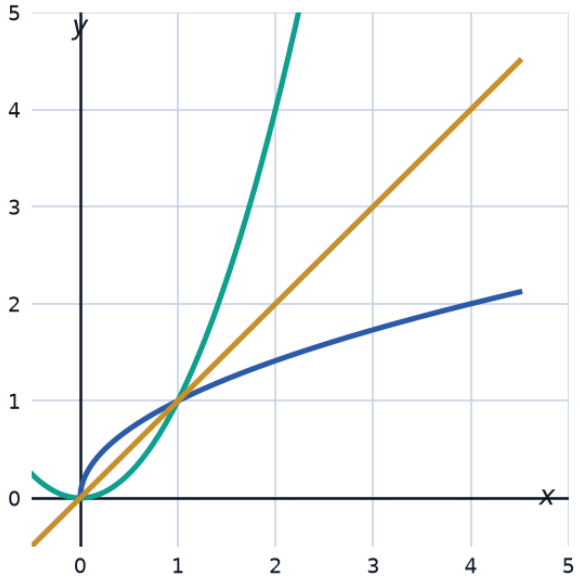
6 [Foundational] • Target: Verifying inverse functions

Find the inverse of $f(x) = 2x^3 + 1$, and verify that $f(f^{-1}(x)) = x$ for one value, say $x = 17$.



7 [Foundational] • Target: Visualizing inverse functions as reflections

The graph shows $f(x) = x^2$ (restricted to $x \geq 0$) and its inverse $f^{-1}(x) = \sqrt{x}$, which are reflections of each other across the line $y = x$.



a State the domain restriction placed on $f(x) = x^2$ needed for it to have an inverse, and explain why.

b Confirm from the graph that f and f^{-1} intersect on the line $y = x$, and find that intersection point algebraically.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Domain and range of composite and inverse functions**

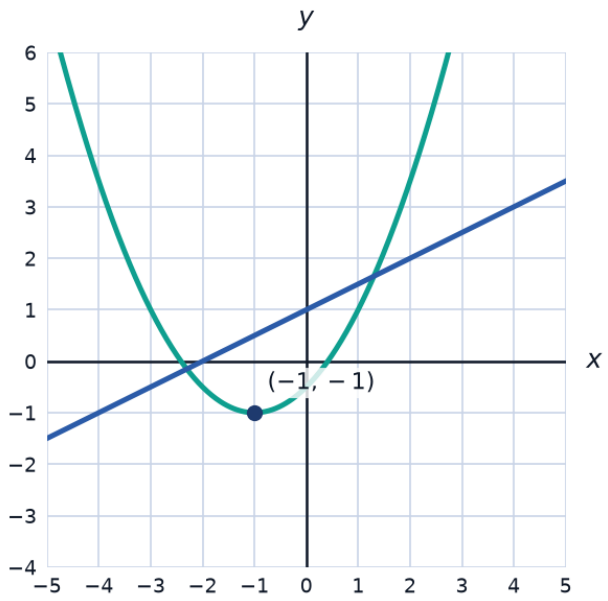
Given $f(x) = \frac{1}{x-3}$ and $g(x) = x + 1$, find the domain of $(f \circ g)(x)$.

9 **[Intermediate]** • **Target: Domain and range of composite and inverse functions**

If f has domain $[1, \infty)$ and range $[0, \infty)$, state the domain and range of f^{-1}.

10 **[Intermediate]** • **Target: Applying composite and inverse functions in context**

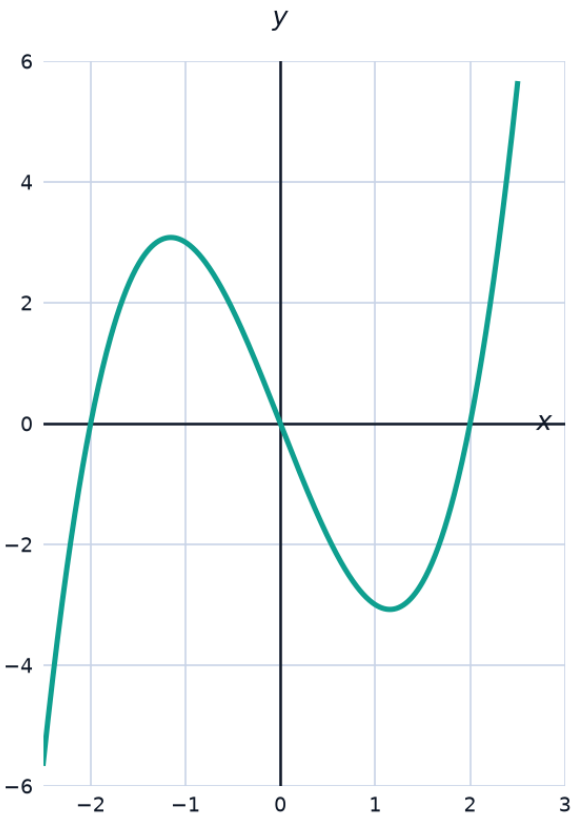
A temperature-conversion function $C(F) = \frac{5}{9}(F - 32)$ converts Fahrenheit to Celsius. Find $C^{-1}(F)$ and interpret what it computes.



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11 [Intermediate] • Target: Using the horizontal line test

Determine whether $f(x) = x^3 - 4x$ has an inverse function over its entire domain. Justify using the horizontal line test.



12 [Intermediate] • Target: Using the horizontal line test

Explain why $f(x) = 5x - 2$ passes the horizontal line test for all real x, and find its inverse.

13 [Intermediate] • Target: Composite functions with restricted domains

Given $f(x) = \frac{1}{x}$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and state its domain.

14 [Intermediate] • Target: Decomposing a composite function

Given $f(x) = 7x + 7$, find $f^{-1}(-21)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

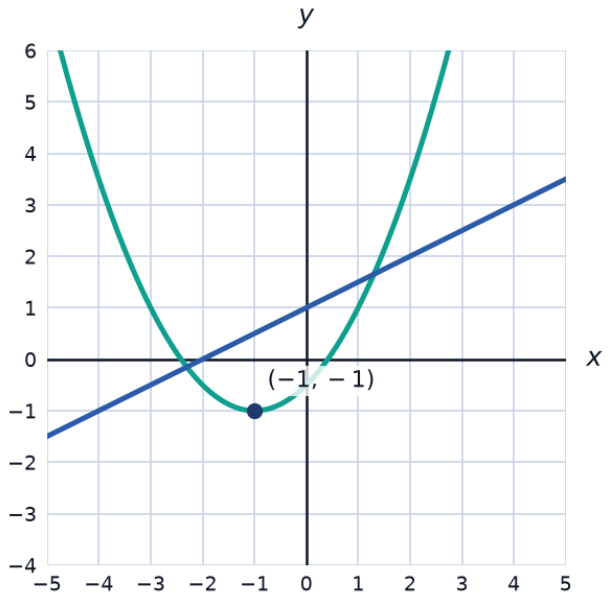
15 [Challenging] • Target: Verifying inverse functions • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Given $f(x) = -5x + 8$, find $f^{-1}(23)$.

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16 [Challenging] • Target: Visualizing inverse functions as reflections • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Given $f(x) = 2x - 9$, find $f^{-1}(-19)$.



17 [Challenging] • Target: Domain and range of composite and inverse functions • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Given $f(x) = 8x - 3$, find $f^{-1}(-3)$.

18 [Challenging] • Target: Applying composite and inverse functions in context • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Given $f(x) = -9x + 4$, find $f^{-1}(4)$.

19 [Challenging] • Target: Using the horizontal line test • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Given $f(x) = 5x - 7$ and $g(x) = -x^2 + 5$, find $(g \circ f)(-3)$.

20 [Challenging] • Target: Composite functions with restricted domains • Cross-Lesson: 4.03 Parametric Equations & Modeling

Given $f(x) = -5x - 8$ and $g(x) = 5x^2 - 6$, find $(g \circ f)(-1)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Composing functions

Given $f(x) = x - 1$ and $g(x) = 5x - 4$, find $(f \circ g)(-6)$.

a. -34

b. -36

c. -35

d. -39

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22 [Foundational] • Target: Decomposing a composite function

Given $f(x) = 5x - 2$ and $g(x) = x + 2$, find $(f \circ g)(-2)$.

a. -3

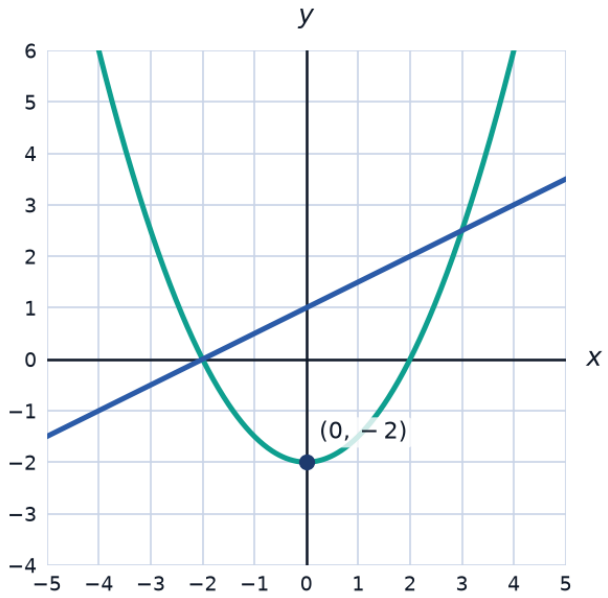
b. -10

c. -2

d. -1

23 [Foundational] • Target: Verifying inverse functions

Given $f(x) = 9x + 5$ and $g(x) = -8x - 9$, find $(f \circ g)(-2)$.



a. 69

b. 67

c. 68

d. 95

24 [Foundational] • Target: Visualizing inverse functions as reflections

Given $f(x) = -9x + 2$ and $g(x) = -8x - 1$, find $(f \circ g)(6)$.

a. 442

b. 444

c. 415

d. 443

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25 [Foundational] • Target: Domain and range of composite and inverse functions

Given $f(x) = 4x - 1$ and $g(x) = -7x - 3$, find $(f \circ g)(0)$.

a. -13

b. -14

c. 4

d. -12

26 [Foundational] • Target: Applying composite and inverse functions in context

Given $f(x) = 6x - 9$ and $g(x) = -x - 1$, find $(f \circ g)(5)$.

a. -46

b. -22

c. -45

d. -44

27 [Foundational] • Target: Using the horizontal line test

Given $f(x) = 6x + 4$ and $g(x) = 9x - 6$, find $(f \circ g)(1)$.

a. 84

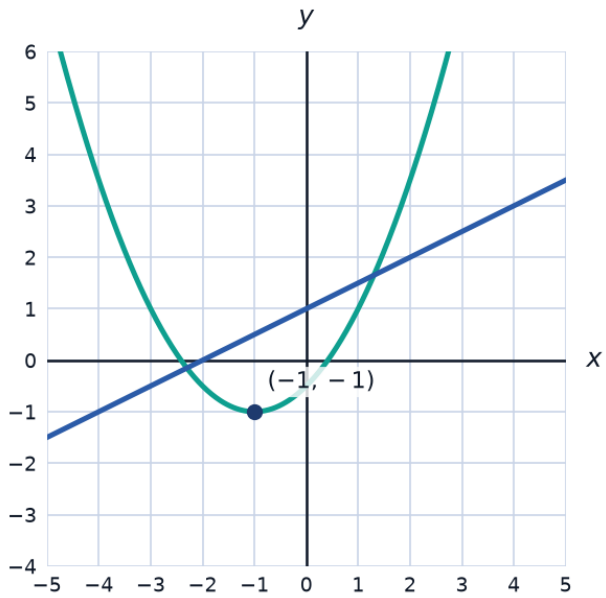
b. 21

c. 23

d. 22

28 [Foundational] • Target: Composite functions with restricted domains

Given $f(x) = -3x - 8$ and $g(x) = 4x - 5$, find $(f \circ g)(1)$.



a. -4

b. -6

c. -5

d. -49

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Composition of Functions and Domain**

Given $f(x) = 9x - 1$, find $f^{-1}(-73)$.

a. -8

b. -73

c. -7

d. -9

30 **[Intermediate]** • **Target: Finding Inverse Functions**

Given $f(x) = -4x + 5$, find $f^{-1}(29)$.

a. -6

b. -7

c. -5

d. 29

31 **[Intermediate]** • **Target: Graphical Symmetry & Invertibility**

Given $f(x) = 2x + 6$, find $f^{-1}(-4)$.

a. -6

b. -5

c. -4

d. -7

32 **[Intermediate]** • **Target: Verifying Inverses Algebraically**

Given $f(x) = -8x + 7$, find $f^{-1}(-25)$.

a. 4

b. 3

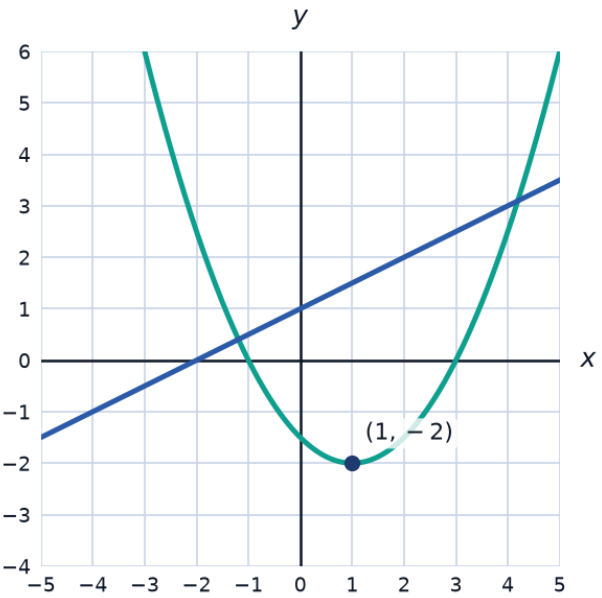
c. -25

d. 5

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33 **[Intermediate]** • **Target: Composing functions**

Given $f(x) = 6x + 7$, find $f^{-1}(-11)$.



a. -11

b. -3

c. -4

d. -2

34 **[Intermediate]** • **Target: Decomposing a composite function**

Given $f(x) = -2x - 1$ and $g(x) = 5x^2$, find $(g \circ f)(1)$.

a. 45

b. 43

c. -15

d. 47

35 **[Intermediate]** • **Target: Verifying inverse functions**

Given $f(x) = -7x$, find the slope and y-intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $(-7, 0)$

b. $\left(-\frac{1}{7}, 0\right)$

c. $\left(\frac{1}{7}, 0\right)$

d. $\left(\frac{5}{14}, 0\right)$

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36 [Intermediate] • Target: Visualizing inverse functions as reflections

Given $f(x) = -9x - 1$, find the slope and y-intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $\left(\frac{1}{9}, -\frac{1}{9}\right)$

b. $(-9, -1)$

c. $\left(-\frac{1}{9}, -\frac{1}{9}\right)$

d. $\left(-\frac{1}{9}, \frac{1}{9}\right)$

37 [Intermediate] • Target: Domain and range of composite and inverse functions

Given $f(x) = 6x - 5$, find the slope and y-intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $\left(-\frac{1}{6}, \frac{5}{6}\right)$

b. $(6, -5)$

c. $\left(\frac{1}{6}, -\frac{5}{6}\right)$

d. $\left(\frac{1}{6}, \frac{5}{6}\right)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applying composite and inverse functions in context • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Determine whether $f(x) = x^3 - 3x$ has an inverse function over its entire domain, using the horizontal line test.

a. Does NOT have an inverse function over its entire domain (not one-to-one)

b. Has an inverse only on the interval $x \geq 0$

c. Cannot be determined without a graph

d. Has an inverse function over its entire domain (strictly increasing)

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39 [Challenging] • Target: Using the horizontal line test • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Determine whether $f(x) = x^3 - 2x$ has an inverse function over its entire domain, using the horizontal line test.

a. Does NOT have an inverse function over its entire domain (not one-to-one)

b. Cannot be determined without a graph

c. Has an inverse function over its entire domain (strictly increasing)

d. Has an inverse only on the interval $x \geq 0$

40 [Challenging] • Target: Composition of Functions and Domain • Cross-Lesson: 4.03 Parametric Equations & Modeling

Given $f(x) = 2x + 5$ and $g(x) = x^2 - 1$, find $(f \circ g)(x)$.

a. $2x^2 + 3$

b. $4x^2 + 20x + 24$

c. $2x^2 + 9$

d. $2x^3 - 1$

41 [Challenging] • Target: Composition of Functions and Domain • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

If $h(x) = (3x - 4)^5$, which two functions f and g satisfy $h(x) = (f \circ g)(x)$?

a. $f(x) = x^5$ and $g(x) = 3x - 4$

b. $f(x) = 3x - 4$ and $g(x) = x^5$

c. $f(x) = 3x^5$ and $g(x) = x - 4$

d. $f(x) = (x - 4)^5$ and $g(x) = 3x$

42 [Challenging] • Target: Finding Inverse Functions • Cross-Lesson: 1.02 Rational Functions & Asymptotes

What is the inverse function of $g(x) = x^3 + 7$?

a. $g^{-1}(x) = \sqrt[3]{x - 7}$

b. $g^{-1}(x) = \sqrt[3]{x} - 7$

c. $g^{-1}(x) = (x - 7)^3$

d. $g^{-1}(x) = \frac{1}{x^3 + 7}$

43 [Challenging] • Target: Graphical Symmetry & Invertibility • Cross-Lesson: 1.03 Rates of Change & Concavity

The graphs of $y = f(x)$ and $y = f^{-1}(x)$ are reflections of each other across which line?

a. $y = x$

b. $y = -x$

c. $x = 0$ (y-axis)

d. $y = 0$ (x-axis)

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44 [Challenging] • Target: Graphical Symmetry & Invertibility • Cross-Lesson: 2.01 Exponential Functions & Growth Models

If $f(2) = 7$ and $f'(2) = 4$, what is $(f^{-1})(7)$?

a. 2

b. 7

c. $\frac{1}{4}$

d. $\frac{1}{2}$

45 [Challenging] • Target: Verifying Inverses Algebraically • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Let $f(x) = 5x - 9$ and $g(x) = \frac{x+9}{5}$. Evaluate $f(g(x))$.

a. x

b. $5x$

c. $x - 9$

d. 1

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